

Task 1 — Interpretability across boundary conditions

all-Dirichlet vs all-Neumann vs mixed, on the unit geometry (1D/2D/3D)

The over-count $\hat{N}$ is geometric (BC-free); the BC only perturbs $\lambda_{\min}$; a coarse space neutralizes it.
(asm_bug_demo/bc_suite.c)

Problem definition — the three concrete examples

$-\Delta u = f$ on the unit hypercube $[0, 1]^d$, $d=1, 2, 3$; $(2d+1)$ -point FD, n nodes/axis

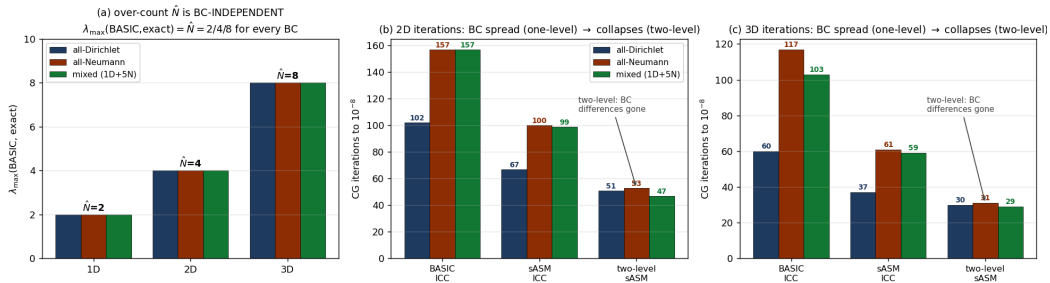
- ▶ **(D) all-Dirichlet:** $u=0$ on every face. SPD. *This is the existing 1D/2D minimal case.*
- ▶ **(N) all-Neumann:** $\partial u / \partial n = 0$ on every face. **Singular** — null space = constants.
- ▶ **(M) mixed:** 1 face Dirichlet + $(2d-1)$ faces Neumann (the 3D Sys 3 type).

Grid: 1D $n=256$ (8 subdomains), 2D 64^2 (4×4), 3D 24^3 ($4 \times 4 \times 4$); overlap $O=2$. BASIC vs sASM additive Schwarz, exact (Cholesky) / ICC(0) subdomain solves, optional Nicolaides two-level; outer CG to 10^{-8} .

Singular case: constant MatNullSpace (mean-zero RHS, nullspace-aware coarse solve).

Item-by-item: what the BC changes, and what it does not

Task 1 — interpretability vs boundary conditions (unit geometry): the over-count is geometric (BC-free); the BC perturbs $\lambda_{\min}$, which the coarse space neutralizes



- ▶ (a) the over-count $\lambda_{\max}(\text{BASIC, exact}) = \hat{N} = 2/4/8$ is **identical for all three BC** — it is geometric.
- ▶ (b,c) one-level iterations differ by BC (2D BASIC-ICC 102/157/157; Neumann & mixed are harder); the **two-level coarse space collapses the spread** (2D sASM 67/100/99 $\rightarrow$ 51/53/47).

The BC only moves $\lambda_{\min}$; the coarse space neutralizes it

Item-by-item summary

	all-Dirichlet	all-Neumann	mixed	BC-dependent?
over-count $\hat{N}$ (2D)	4	4	4	no (geometric)
$\lambda_{\max}$ (BASIC,exact)	$= \hat{N}$	$= \hat{N}$	$= \hat{N}$	no
$\lambda_{\min}$ 1-level (2D)	0.0071	0.0039	0.0010	yes
1-level sASM iters (2D)	67	100	99	yes
$\lambda_{\min}$ two-level (2D)	0.045	0.053	0.043	no (aligned)
two-level iters (2D)	51	53	47	no (collapsed)

- ▶ The BC changes the **global/constant mode**: Neumann is *singular* (constant null space); mixed is barely pinned by its one Dirichlet face (smallest $\lambda_{\min}$); Dirichlet is best.
- ▶ **The Nicolaides PoU coarse space spans the constant** ($\sum_i \phi_i = 1$) — exactly the mode the BC perturbs. So it lifts *and aligns* $\lambda_{\min}$ across BC. For all-Neumann the coarse space *is* the null space $\Rightarrow$ the perfect fix.

Conclusion (Task 1)

- ▶ **Over-count is BC-free.** $\hat{N}$ and $\lambda_{\max}(\text{BASIC,exact}) = \hat{N}$ are identical for all-Dirichlet, all-Neumann and mixed — so the entire sASM “remove $\hat{N}$ ” story carries over unchanged.
- ▶ **The BC only perturbs $\lambda_{\min}$** (the global/constant mode): Neumann singular, mixed near-singular, Dirichlet best $\Rightarrow$ one-level iterations carry a BC spread.
- ▶ **A coarse space neutralizes the BC.** Nicolaides (spans the constant) collapses the $\lambda_{\min}$ spread and the iteration spread; for all-Neumann it is the *exact* coarse space (= the null space).